

C3.1 How are the atoms held together in a metal?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Chemists use a model of metal structure to explain the properties of metals (1aS3). In the model, metal atoms are arranged closely together in a giant structure, held together by attraction between the positively charged atoms and a 'sea' of negatively charged electrons. Metals are malleable and ductile because the ions can slide over each other but still be held together by the electrons; they conduct electricity and heat because their electrons are free to move; and they have high boiling points and melting points due to the strong electrostatic attraction between metal ions and the electrons. These properties of metals make them useful.	1. describe the nature and arrangement of chemical bonds in metals 2. explain how the bulk properties of metals are related to the type of bonds they contain

Linked learning opportunities

Ideas about Science:

- use the model of metal structure to explain properties of metals (1aS3)
